

Quiz (Habanero)

- Q1.** A weather station records 15 inches of rainfall in 3 hours. What is the rate of rainfall per hour?
(A) 3 inches/hour
(B) 5 inches/hour
(C) 6 inches/hour
(D) 8 inches/hour
(E) 10 inches/hour
- Q2.** If y varies directly with x , and $y = 12$ when $x = 4$, find the value of y when $x = 6$.
(A) 12
(B) 16
(C) 18
(D) 20
(E) 24
- Q3.** A ship travels 240 miles in 4 hours. What is its average speed?
(A) 40 miles/hour
(B) 50 miles/hour
(C) 60 miles/hour
(D) 70 miles/hour
(E) 80 miles/hour
- Q4.** The temperature in a city decreases by 2°ircC for every 1000 feet of elevation. If the temperature at sea level is 25°ircC , find the temperature at an elevation of 3000 feet.
(A) 19°ircC
(B) 21°ircC
(C) 23°ircC
(D) 25°ircC
(E) 27°ircC
- Q5.** A water tank can be filled by a pump at a rate of 2 cubic meters per minute. How many minutes will it take to fill a tank that holds 30 cubic meters of water?
(A) 10 minutes
(B) 12 minutes
- (C) 15 minutes
(D) 18 minutes
(E) 20 minutes
- Q6.** A group of friends want to share some candy equally. If they have 48 pieces of candy and there are 8 friends, how many pieces will each friend get?
(A) 4 pieces
(B) 5 pieces
(C) 6 pieces
(D) 7 pieces
(E) 8 pieces
- Q7.** A cyclist travels 25 miles in 2 hours. What is the cyclist's average speed in miles per hour?
(A) 10 miles/hour
(B) 12 miles/hour
(C) 12.5 miles/hour
(D) 15 miles/hour
(E) 20 miles/hour
- Q8.** The cost of a ticket to a concert is directly proportional to the number of tickets purchased. If 2 tickets cost 20, how much will 5 tickets cost?
(A) 25
(B) 30
(C) 40
(D) 45
(E) 50

Open Ended Questions (FRQ)

- Q9.** A company is planning to build a new wind farm. The cost of the wind farm is directly proportional to the number of wind turbines. If the cost of 3 wind turbines is 900,000, find the cost of 7 wind turbines. Show your work and explain your reasoning.
- Q10.** A city's water reservoir can hold 1,200,000 gallons of water. Due to a drought, the reservoir is being filled at a rate of 5,000 gallons per hour. How many hours will it take to fill the reservoir? If the city's population is growing at a rate of 2

Chef's Tips

- Calculate the unit rate by dividing the total amount by the total time or quantity.
- Use the formula $y = kx$ for direct variation, where k is the constant of variation.
- Use the formula $y = \frac{k}{x}$ for inverse variation, where k is the constant of variation.
- Chef's Tips